

# KSAC GatePass & Society Room Booking System

## Complete Operational Guide for Students, Wardens, KSAC Authorities & Administrators

A progressive web application engineered for KIIT University to digitize hostel curfew extensions, manage 31+ official student societies, automate central room reservations, and enforce real-time multi-gate security corroboration.

|                                                                 |                                                            |
|-----------------------------------------------------------------|------------------------------------------------------------|
| <b>INSTITUTION</b><br>KIIT Deemed to be University              | <b>SYSTEM VERSION</b><br>v2.4.0 (PWA Mobile-First Edition) |
| <b>PRIMARY URL</b><br>http://localhost:3000 (Production Portal) | <b>SCOPE</b><br>31 Societies • 15+ Hostels • Central Desk  |

# 1. System Overview & 6-Step GatePass Protocol

The KSAC GatePass platform operates on a synchronized state machine connecting three physical nodes: the **Resident Hostel**, the **KSAC Central Activity Center**, and the **University Administration**.

## Core Security Principle: Dual-Corroborated Transit

GatePass requests are verified by KSAC authorities first before hostel departure, and student returns are verified by the resident hostel warden. This guarantees zero unauthorized curfew extensions while preserving student freedom for legitimate extracurricular activities.

## The 6-Step Realigned GatePass Flow

### 1 Step 1: Student Submits GatePass Request

The student selects their designated society, purpose, and target return time (e.g. 09:30 PM, 10:00 PM). The pass is created with status **PENDING** and linked to the student's resident hostel.

### 2 Step 2: KSAC Authority Queue Notification

The KSAC Desk receives the pending request with the student's name, roll number, resident hostel, and society activity details in the central queue.

### 3 Step 3: KSAC Verification & Acceptance

KSAC Officer reviews and clicks "**Verify & Accept**". The pass transitions to **APPROVED**, and the KSAC arrival/session timestamp is officially stamped.

### 4 Step 4: Warden Real-Time Outside Tracking

The Resident Warden's dashboard automatically reflects that the student is outside at KSAC with an approved in-time extension curfew.

### 5 Step 5: Warden Confirms Hostel Return

When the student returns to the hostel gate, the Warden clicks "**Confirm Hostel Entry**". The pass transitions to **RETURNED** with the verified return timestamp.

### 6 Step 6: KSAC Authority Corroboration

The KSAC Desk updates in real-time to show that the student has safely returned to their hostel, completing the transit audit trail.

## 2. User Manual by Role

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### A. Student User Guide (Mobile & Desktop)

Students access their dedicated **Student Hub** designed natively for smartphones as a Progressive Web App (PWA).

- **PWA Installation:**
  - **iOS (iPhone):** Open Safari → tap Share icon → select "*Add to Home Screen*".
  - **Android:** Open Chrome → tap the install prompt or menu → "*Install App*".
- **Requesting a GatePass:** Navigate to the *GatePass* tab. Select your society/wing, select or enter your purpose, tap one of the quick return time chips (06:00 PM, 08:30 PM, 09:30 PM, 10:00 PM), and tap "**Request GatePass**".
- **Active Boarding Pass Screen:** Displays live transit state, resident hostel badge, society, target curfew, and real-time status instructions.
- **Society Room Booking (Presidents & Vice-Presidents):**
  - Society leaders have unlocked access to the *Society Room Booking* tab.
  - Select booking date, start time, and end time using one-tap duration chips (+1h, +1.5h, +2h, +3h, Until 7 PM).
  - System automatically validates booking hours (10:00 AM - 07:00 PM, min 1 hour) and prevents room slot overlaps.

### B. Hostel Warden Control Center Guide

Wardens monitor curfew compliance and verify physical hostel check-ins.

- **Students Outside at KSAC:** View all students from your hostel currently at KSAC with verified extension curfews.
- **Confirming Hostel Entry:** When a student presents their ID at the hostel gate, click "**Confirm Hostel Entry**". This stamps the arrival time and notifies KSAC authorities immediately.
- **Hostel Filter:** Chief Wardens can switch between specific hostels (e.g. KP-7A, KP-6, QC-2) or view all resident hostels.

### C. KSAC Central Authority Desk Guide

KSAC Officers supervise university-wide society sessions, verify passes, and manage room allocations.

- **Pending GatePass Verification Queue:** Review incoming requests from students, check society credentials, and click "**Verify & Accept**" or "**Reject**".
- **Hostel Return Tracking:** Live stream showing students who have safely checked back into their hostels with warden-verified timestamps.
- **Room Approvals Desk:** Review and approve/reject society room reservations with collision detection against approved slots.

### D. Master Admin Control Center Guide

Master Administrators (admin@kiit.ac.in) possess root administrative authority over all entities.

#### Full CRUD Administrative Privileges:

- **Students & Users:**  
Add new students/wardens, edit profiles, assign hostels, or delete accounts.
- **Room Booking Access:**  
Toggle room booking permissions, assign societies, and designate studio rooms.
- **GatePass Overrides:**  
Force-approve, force-return, or delete any gatepass record.
- **Room Reservations:**  
Manage, override, or delete any room reservation.
- **Audit Trail:**  
Inspect real-time audit logs of all gate movements and administrative actions.

### 3. Default Accounts & Credentials Reference

All test accounts are configured with the universal password: password123

| ROLE / PERSONA    | OFFICIAL EMAIL      | DEFAULT HOSTEL     | PRIVILEGES & SCOPE                                 |
|-------------------|---------------------|--------------------|----------------------------------------------------|
| MASTER ADMIN      | admin@kiit.ac.in    | KIIT Central Admin | Full CRUD on Users, Passes, Bookings, System Stats |
| KSAC DESK OFFICER | ksac@kiit.ac.in     | KSAC Central       | Verify Passes, Approve Rooms, Return Tracking      |
| CHIEF WARDEN      | warden@kiit.ac.in   | ALL / KP-7A        | Outside Students Tracking, Confirm Hostel Entry    |
| FILM SOCIETY LEAD | 23051959@kiit.ac.in | KP-7A              | President (KIIT Film Society), Room 102 Booking    |
| KORUS PRESIDENT   | 2330448@kiit.ac.in  | KP-6               | President (KORUS), Room 203 Audio Studio           |
| GENERAL STUDENT   | 21051234@kiit.ac.in | KP-7A              | GatePass Requests, In-Time Extensions              |
| GENERAL STUDENT   | 23058888@kiit.ac.in | QC-2               | Queen's Castle Resident GatePasses                 |

### 4. Technical Specifications & Security Safeguards

- Masked Gateway & Anti-Replay Protection:** All client-server communication passes through /api/gateway using opaque opcodes and 60-second cryptographic nonces.
- Multi-Model Rate Limiting:** Sliding window rate limiters protect authentication endpoints against brute force attempts.
- Responsive & PWA Shell:** Service worker caches static assets with stale-while-revalidate for instantaneous offline loads.
- Zero Horizontal Overflow:** Viewport tested on 375px (iPhone SE), 393px (iPhone 15 Pro), 412px (Pixel 7), 768px (iPad), and 1440px (Desktop).

**Technical Support & Bug Reporting:**

For support, server maintenance, or society room configuration inquiries, contact the KSAC Digital Infrastructure Team at admin@kiit.ac.in.